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Milestone Two: Software Design and Engineering Enhancement

1. What is the artifact, and when was it created?

This is the CS-320 Contact, Task, and Appointment work from summer 2025 (CS-320 Software Testing, Automation, and QA, 2025 C-4). The original files are in the original/ folder of this zip and at https://github.com/AmauriHopewell/CS320. For this milestone I enhanced Contact. Contact.java still stores an ID, first name, last name, phone, and address. The ID is final. The constructor and setters still reject nulls, names longer than 10 characters, addresses longer than 30, and phones that are not ten digits. ContactService still adds, updates, deletes, and looks up by ID. I wrote those rules in CS-320 with JUnit. This week's work is the layer around that service, not a rewrite of the entity.

2. Why this artifact, and how was it improved?

I picked Contact because it already had the field rules and tests, and it was still a prototype. State died when the process ended. There was no HTTP API. There was no login, even though the records are personal. The README also said HashMap while ContactService walked an ArrayList. The Module One plan was to keep ContactService as the business layer and put a repository, SQLite, a small REST controller, an auth filter, and an audit log around it (Hopewell, 2026). That is what is in enhanced/.

ContactRepository is the new storage boundary. InMemoryContactRepository is a HashMap. Unit tests use that, so lookup is O(1) and the original Contact / ContactService cases still run. SqliteContactRepository is the production store. A test writes a contact, opens a new service on the same file, and reads it back. ContactApi is a JDK HttpServer. I did not use Spring Boot. The plan said a full framework would bury the design, and I wanted ContactApi.java to be readable in one sitting (Hopewell, 2026; Southern New Hampshire University [SNHU], n.d.). AuthService has two users, staff and admin. Passwords are salted SHA-256. They come from CONTACT_STAFF_PASSWORD and CONTACT_ADMIN_PASSWORD. App.main exits if those are missing, so there is no default password in source. A missing token returns 401 with {}. A staff token on POST, PUT, or DELETE returns 403 with {} and no contact body. AuditLog records actor, action, id, and status. It does not take a phone number or address as arguments. I also named the 10 / 30 limits as constants and removed the System.out.println calls from getContact. mvn test is 37 tests, all green.

3. Course outcomes from Module One

The Module One plan said this enhancement would show outcomes 3, 4, and 5. That is still the plan. Outcome 3 is the store trade-off: HashMap for fast tests, SQLite so contacts survive a restart. Outcome 4 is JUnit, a repository, SQLite JDBC, and java.util.logging used on one path. Outcome 5 is the auth filter and the log that skips PII (Hopewell, 2026). Outcome 2 is this write-up. Outcome 1 is the GitHub history after I take comments and push the enhanced tree. I am not changing the coverage map. Task and Appointment still copy the same CRUD. I left them alone this week so Contact would actually finish. The 3-2 directions said to wait for instructor comments before putting this on the ePortfolio (SNHU, n.d.), so Pages still links the original CS-320 repo.

4. What I learned, and what got in the way

The awkward part was moving storage without breaking the CS-320 tests. ContactService used to own an ArrayList. I gave it a ContactRepository and a no-arg constructor that still builds the in-memory map. The old tests construct new ContactService() and do not know about SQLite. The SQLite tests use a temp file from JUnit.

Auth was picky in a useful way. It is easy to return 403 with a JSON error that still includes the contact. I wrote a test that fails if that body contains the sample phone number. Login is POST /login. Tokens are random hex, not the password. OWASP's review guide is why the constructor does not get a default password the way my CS-340 module still does (OWASP Foundation, n.d.).

HttpServer has no path parameters. ContactApi parses /contacts/{id} itself. That is ugly, and it is also the whole HTTP layer in one class, which was the point. The original zip still has the Eclipse packages Contact, com, and Src. The enhanced tree uses edu.snhu.contact. I did not try to "fix" the original packages in place, because this submission has to include the original files as they were.

References

Hopewell, A. (2026, September 6). CS-499 Module One assignment: ePortfolio enhancement plan [Unpublished manuscript]. Southern New Hampshire University.

OWASP Foundation. (n.d.). OWASP code review guide. https://owasp.github.io/www-project-code-review-guide/

Southern New Hampshire University. (n.d.). CS 499 milestone two guidelines and rubric: Enhancement one software design/engineering.

Acknowledgment of generative AI use

I used Grok (xAI) in September 2026 to help organize this narrative against the Milestone Two prompts and to format the Word file. The code in enhanced/ is the Contact service from my CS-320 work plus the layers listed above. I ran mvn test before keeping the test counts in this document.